

1. For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = 2(a - b)x_1x_2 + a(x_1^2 + x_2^2).$$

- (a) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A . (4 marks)

Solutions: (2 marks) The matrix of coefficients is:

$$A = \begin{pmatrix} a & a - b \\ a - b & a \end{pmatrix}$$

(2 marks) Its second order principal minor is equal to $b(2a - b)$. Its first order principal minors are equal to a .

- (b) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all the values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*? (12 marks)

Solutions: (2 marks) A quadratic form is indefinite if $Q(\mathbf{x}) > 0$ and $Q(\mathbf{y}) < 0$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n \setminus \{0\}$. A quadratic form is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$. A quadratic form is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n \setminus \{0\}$.

(4 marks) $Q(\mathbf{x})$ is positive semidefinite if all principal minors are positive which requires:

$$2a \geq b \geq 0.$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(4 marks) $Q(\mathbf{x})$ is negative semidefinite if all principal minors suitably alternate in sign which requires:

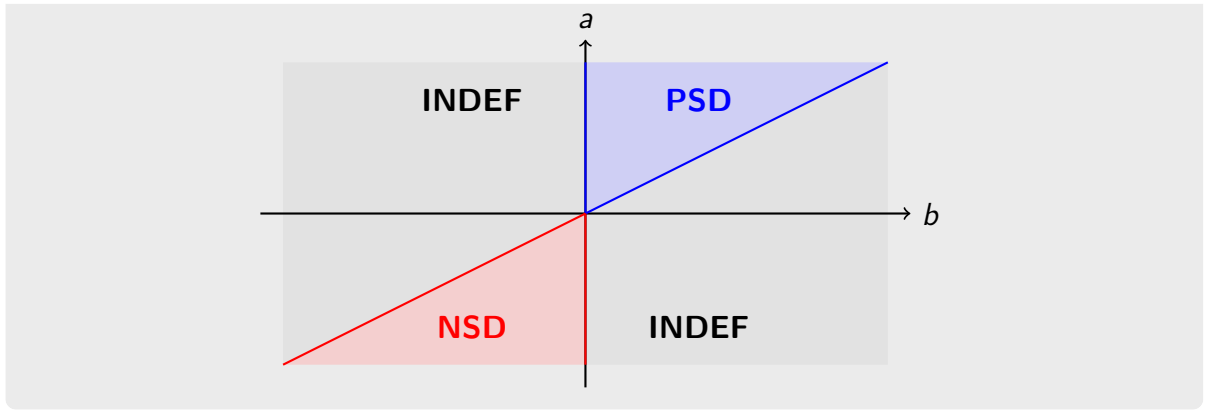
$$2a \leq b \leq 0.$$

The condition accounts for scenarios in which some of the parameters are equal to 0.

(2 marks) $Q(\mathbf{x})$ is indefinite whenever it is not semidefinite.

- (c) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$. (4 marks)

Solutions: (4 marks) The following plot depicts parameter values for which the quadratic form can be signed.



2. Consider the function $f(\mathbf{x}) = x_1^2 x_2 e^{-x_2}$ defined on some convex set U in $\mathbb{R}_{++}^2$.

(a) Define what it means for f to be *quasi-concave* and *quasi-convex*. (6 marks)

Solution: The function f is: (3 marks) *quasi-concave* on U if $C_k^+ = \{\mathbf{x} \in U \mid f(\mathbf{x}) \geq k\}$ is a convex set for every real number k in the image of f ; (3 marks) *quasi-convex* on U if $C_k^- = \{\mathbf{x} \in U \mid f(\mathbf{x}) \leq k\}$ is a convex set for every real number k in the image of f .

(b) Is f *quasi-concave* or *quasi-convex*? Use counterexamples to show that a property fails. (14 marks)

Solution:

(7 marks) The function f is quasi-concave. To show this, consider

$$g(x_1, x_2) = \log(x_1^2 x_2 e^{-x_2}) = 2 \log x_1 + \log x_2 - x_2.$$

The function g is concave since $2 \log x_1$ is concave for $x_1 > 0$, $\log x_2$ is concave for $x_2 > 0$, $-x_2$ is concave (linear), and the sum of concave functions is concave.

Since g is concave and $f(x_1, x_2) = e^{g(x_1, x_2)}$, and since the exponential function is a strictly increasing transformation, f is quasi-concave.

(7 marks) The function f is not quasi-convex. To show this with a counterexample, consider the points $\mathbf{x} = (1, 1/2)$ and $\mathbf{y} = (1, 3/2)$. Observe that

$$\max\{f(\mathbf{x}), f(\mathbf{y})\} < 1/e,$$

where the inequality holds since the function $x_2 e^{-x_2}$ has a unique global maximizer at $x_2 = 1$. By construction, both points must belong to the lower contour set $C_{\max\{f(\mathbf{x}), f(\mathbf{y})\}}^-$.

However, consider the midpoint $\mathbf{z} = (\mathbf{x} + \mathbf{y})/2 = (1, 1)$. At such a point,

$$f(\mathbf{z}) = 1/e.$$

Thus, $\mathbf{z} \notin C_{\max\{f(\mathbf{x}), f(\mathbf{y})\}}^-$, thereby violating quasi-convexity.

3. Consider the problem:

$$\max_{x,y} xy^2 \text{ subject to } x + 2y \leq a, \ x \geq \frac{1}{a}, \ y \geq 0.$$

- (a) For what values of $a > 0$ is Slater condition satisfied? For such values of a , do Kuhn-Tucker necessary first order conditions identify a solution to the constrained optimization problem? **(10 marks)**

Solution: (5 marks) To check whether Slater condition holds, we need a point (x, y) with $x > \frac{1}{a}$, $y > 0$, and $x + 2y < a$. The binding case gives us:

$$\frac{1}{a} + 2 \cdot 0 < a \Rightarrow \frac{1}{a} < a \Rightarrow a^2 > 1 \Rightarrow a > 1.$$

So Slater condition is satisfied for $a > 1$.

(5 marks) For $a > 1$, Kuhn-Tucker necessary conditions identify a solution to the constrained optimization problem, since all constraints are linear (hence convex) and the constraint qualification (Slater condition) holds; moreover $f(x, y) = xy^2$ is quasi-concave on $\mathbb{R}_{++}^2$ (it is a Cobb-Douglas-type function), so the KT necessary conditions are in fact also sufficient for a global maximum.

- (b) Write down the Kuhn-Tucker necessary first order conditions that must be satisfied by the solution of the constrained optimization problem. **(6 marks)**

Solution: (6 marks) The Lagrangian is $\mathcal{L}(x, y|\lambda) = xy^2 - \lambda_1(x + 2y - a) - \lambda_2(-x + 1/a) - \lambda_3(-y)$. The necessary first order conditions are:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= y^2 - \lambda_1 + \lambda_2 = 0, & \frac{\partial \mathcal{L}}{\partial y} &= 2xy - 2\lambda_1 + \lambda_3 = 0, \\ \lambda_1(x + 2y - a) &= 0, & \lambda_2\left(-x + \frac{1}{a}\right) &= 0, & \lambda_3(-y) &= 0, \\ x + 2y - a &\leq 0, & x - \frac{1}{a} &\geq 0, & y &\geq 0, \\ \lambda_1, \lambda_2, \lambda_3 &\geq 0. \end{aligned}$$

- (c) Solve the constrained optimization problem for all values of a satisfying the restrictions found in part (a). **(18 marks)**

Solution: (6 marks) First observe that $y > 0$ at the optimum when $a > 1$. This follows because $f(x, 0) = 0$ for any $x \geq 1/a$, while $f(\frac{1}{a}, \frac{a-1/a}{2}) > 0$ and $(\frac{1}{a}, \frac{a-1/a}{2})$ is feasible (it lies on the budget line with $y > 0$, which is positive precisely because $a > 1$). We can therefore conclude that $\lambda_3 = 0$ at the optimum. Moreover, $x + 2y = a$ at the optimum, because the objective function is strictly increasing in both variables for $x, y > 0$. Thus $\lambda_1 > 0$.

Now, consider the two remaining cases for $a > 1$:

Case 1: (6 marks) $\lambda_1 > 0$, $\lambda_2 = 0$, $\lambda_3 = 0$. By complementary slackness, we have

$x + 2y = a$, $x > \frac{1}{a}$, $y > 0$. From the FOCs, we further find:

$$\begin{aligned} y^2 - \lambda_1 &= 0 \Rightarrow \lambda_1 = y^2, \\ 2xy - 2\lambda_1 &= 0 \Rightarrow \lambda_1 = xy. \end{aligned}$$

Therefore $y^2 = xy$, which gives $x = y$. Substituting into the budget constraint: $x + 2x = a$, so $x = y = \frac{a}{3}$. This solution is valid when $\frac{a}{3} > \frac{1}{a}$, which requires $a > \sqrt{3}$.

Case 2: (9 marks) $\lambda_1 > 0$, $\lambda_2 > 0$, $\lambda_3 = 0$. By complementary slackness for the two binding constraints, we have $x = \frac{1}{a}$, and $y = \frac{a - 1/a}{2} = \frac{a^2 - 1}{2a}$. From the FOCs, we further find:

$$\lambda_1 = xy = \frac{a^2 - 1}{2a^2}, \quad \lambda_2 = \lambda_1 - y^2 = \frac{a^2 - 1}{2a^2} - \left(\frac{a^2 - 1}{2a}\right)^2 = \frac{(a^2 - 1)(3 - a^2)}{4a^2}.$$

Since $a > 1$ gives $a^2 - 1 > 0$, we need $\lambda_2 \geq 0 \iff 3 - a^2 \geq 0 \iff a \leq \sqrt{3}$.

Summarizing: $(x^*, y^*) = \left(\frac{1}{a}, \frac{a^2 - 1}{2a}\right)$ for $1 < a \leq \sqrt{3}$, and $(x^*, y^*) = \left(\frac{a}{3}, \frac{a}{3}\right)$ for $a \geq \sqrt{3}$.

- (d) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \lambda(a))$ found in part (c). (6 marks)

Solution: (6 marks) Since both the budget constraint and the constraint $x \geq \frac{1}{a}$ depend explicitly on a , the Envelope Theorem gives:

$$\frac{dv(a)}{da} = \frac{\partial \mathcal{L}(x, y | \lambda, a)}{\partial a} \Big|_{(x, y, \lambda) = (x(a), y(a), \lambda(a))} = \lambda_1(a) + \frac{\lambda_2(a)}{a^2}.$$

Case 1: For $a \geq \sqrt{3}$, $\lambda_2 = 0$ and $\lambda_1 = y^2 = a^2/9$, so

$$\frac{dv(a)}{da} = \frac{a^2}{9}.$$

Case 2: For $1 < a \leq \sqrt{3}$, $\lambda_1 = \frac{a^2 - 1}{2a^2}$ and $\lambda_2 = \frac{(a^2 - 1)(3 - a^2)}{4a^2}$, so

$$\frac{dv(a)}{da} = \frac{a^2 - 1}{2a^2} + \frac{1}{a^2} \cdot \frac{(a^2 - 1)(3 - a^2)}{4a^2} = \frac{(a^2 - 1)(a^2 + 3)}{4a^4}.$$

The two expressions agree at $a = \sqrt{3}$, confirming continuity of the value function's slope. Summarizing:

$$\frac{dv(a)}{da} = \begin{cases} \frac{(a^2 - 1)(a^2 + 3)}{4a^4}, & 1 < a \leq \sqrt{3} \\ \frac{a^2}{9}, & a \geq \sqrt{3} \end{cases}$$

4. Consider the correspondence $\mathbf{h} : [0, 4]^2 \rightrightarrows [0, 4]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} [0, 2] & \text{for } x_2 \in [0, 1) \\ [0, 4] & \text{for } x_2 = 1 \\ [2, 4] & \text{for } x_2 \in (1, 4] \end{cases}$$

$$h_2(\mathbf{x}) = 4 - x_1 \text{ for } x_1 \in [0, 4].$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence. **(20 marks)**

Solution: (4 marks) The correspondence h_1 has non-empty, compact and convex values (closed intervals), and it is upper-hemicontinuous: at the only point where its definition changes discontinuously, $x_2 = 1$, the value $h_1(\cdot, 1) = [0, 4]$ equals the union of the neighbouring values $[0, 2]$ (for $x_2 \uparrow 1$) and $[2, 4]$ (for $x_2 \downarrow 1$), so the graph of h_1 is closed. The function h_2 is continuous (hence upper-hemicontinuous with singleton, trivially convex values). The domain $[0, 4]^2$ is compact and convex, and $\mathbf{h}$ maps it into itself. Therefore Kakutani's Fixed Point Theorem applies, guaranteeing at least one fixed point.

(16 marks) For a fixed point, we need $(x_1, x_2) \in \mathbf{h}(x_1, x_2) = h_1(x_1, x_2) \times h_2(x_1, x_2)$.

Since $h_2(x_1, x_2) = 4 - x_1$, at any fixed point we must have $x_2 = 4 - x_1$, i.e. $x_1 = 4 - x_2$. Given $x_1 \in [0, 4]$, this gives $x_2 \in [0, 4]$ automatically. Now we check when $x_1 = 4 - x_2 \in h_1(x_1, x_2)$:

Case 1: If $x_2 \in [0, 1)$, then $h_1(\mathbf{x}) = [0, 2]$ and $x_1 = 4 - x_2 \in (3, 4]$. Since $(3, 4] \cap [0, 2] = \emptyset$, there are no solutions in this case.

Case 2: If $x_2 = 1$, then $x_1 = 4 - x_2 = 3$, and $h_1(\mathbf{x}) = [0, 4] \ni 3$. Thus $(3, 1)$ is a fixed point.

Case 3: If $x_2 \in (1, 4]$, then $h_1(\mathbf{x}) = [2, 4]$ and $x_1 = 4 - x_2 \in [0, 3)$. Since $[0, 3) \cap [2, 4] = [2, 3)$, any $x_1 \in [2, 3)$ (with $x_2 = 4 - x_1 \in (1, 2]$) gives a fixed point $(x_1, 4 - x_1)$.

Combining Cases 2 and 3, the set of fixed points is the line segment

$$\{(x_1, x_2) \in [0, 4]^2 : x_2 = 4 - x_1, 2 \leq x_1 \leq 3\},$$

running from $(2, 2)$ to $(3, 1)$.

